CYBERCRIME CASE STUDY

# Phishing Attack

How a college student lost money to a fake university email

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| ~ | #1 | +75% |
|---|---|---|
| **4.7M+** | **Most Common** | **Rise in India** |
| Phishing attacks in 2022 (APWG Report) | Initial attack vector globally (IBM 2023) | Education sector phishing (CERT-In 2022) |

## 01 What is Phishing?

### Definition

Phishing is a cybercrime where an attacker **impersonates a trusted entity** — a bank, university, or government body — to trick victims into revealing sensitive information such as passwords, OTPs, or bank details.

The name comes from fishing — casting a wide net to lure unsuspecting victims. It exploits human psychology rather than technical vulnerabilities, using fear, urgency, and trust as its primary weapons.

**DELIVERED VIA**

- Email (most common)
- SMS — called Smishing
- Voice calls — called Vishing
- Fake websites / QR codes

## 02 The Case: Fake University Portal Attack

### Background

This is a **realistic fictional scenario** based on documented phishing patterns reported by CERT-In and the National Cyber Crime Reporting Portal. It reflects attack patterns widely reported among college students across India.

### The Victim

| NAME | YEAR | INSTITUTION | TRIGGER |
|---|---|---|---|
| **Arjun Mehta** | 2nd Year B.Tech | Engineering University, M.P. | Upcoming semester exams |

## The Phishing Email Arjun Received

| | |
|---|---|
| FROM: | VIT Helpdesk – IT Services <support@vitbhopal-helpdesk.in> |
| TO: | arjun.mehta@student.vitbhopal.ac.in |
| SUBJECT: | **URGENT: Your Student Portal Access Will Be Suspended in 24 Hours** |
| BODY: | Dear Student, Our portal has been upgraded. All students must re-verify login credentials before midnight to avoid losing access to exam results, attendance, and fee records. **Click Here to Verify Now** |

**Red flag ignored:** The sender domain was vitbhopal-helpdesk.in — NOT the official vitbhopal.ac.in. Arjun saw the display name 'VIT Helpdesk' and did not check further.

## 03 How the Attack Unfolded — Step by Step

**1 Reconnaissance — Gathering Information**
The attacker visited the university's official website and noted its logo, colour scheme, and department names. Student email addresses were scraped from publicly available notice boards and event pages — all done without any hacking.

**2 Registering a Fake Domain**
A domain nearly identical to the university's was registered: vitbhopal-helpdesk.in instead of vitbhopal.ac.in. Cost: under Rs. 500. Time to go live: under 2 hours. The domain also had a valid HTTPS certificate, displaying the 'padlock' icon in the browser.

**3 Building the Fake Login Page**
A pixel-perfect copy of the student portal login page was created. Same logo, same layout, same form fields — the only difference was the URL. Most users never look at the address bar.

**4 Sending the Phishing Email**
A mass email was sent to hundreds of student addresses simultaneously. Key psychological triggers: URGENCY ('24 hours'), FEAR ('lose exam results'), AUTHORITY ('IT Helpdesk'), and FAMILIARITY (referenced upcoming exams to feel personal and credible).

**5 Arjun Clicks the Link**
Alarmed by the email, Arjun clicked the link and entered his student ID and password on the fake portal. The page showed 'Verification successful' — and he closed the tab, completely unaware that his credentials had just been stolen.

**6** **Credential Harvesting**
Arjun's username and password were instantly transmitted to the attacker's server. Within minutes, the attacker logged into the real student portal and extracted: full name, date of birth, registered mobile number, home address, and linked email.

**7** **Escalation — UPI Fraud**
The attacker accessed Arjun's Gmail (same password reused). They initiated a UPI password reset which sent an OTP to his phone. Posing as 'IT Support', they called Arjun and asked for the OTP to 'complete portal verification.' Arjun shared it. The attacker reset his UPI PIN and transferred funds out of his bank account within seconds.

## 04 Who is Usually Targeted?

* **College students**
Rely on online portals for exams, attendance, fees — easy to threaten.

$ **Banking customers**
FakeKYC alerts, creditcard verification, and transaction fraud emails.

# **First/second-year students**
Lesscybersecurityexperience;more trusting of official-looking emails.

@ **Government service users**
FakeAadhaar / PAN / incometax refund notifications.

> **Job seekers**
Targeted with fake offer letters, interview calls, and HR portal logins.

+ **Small business owners**
Fake GST notices,vendorpayment requests, and invoice fraud.

**CERT-In Annual Report 2022:** The education sector saw a **75% increase** in phishing incidents targeting institutional email systems, with students as the primary victims.

## 05 Consequences of the Attack

### For the Victim

**Financial loss** — Immediate unauthorised UPI transfers. Recovery is difficult and often only partial.

**Identity theft** — Personal data stolen — could be used to take loans, apply for SIMs, or create fake profiles.

**Academic disruption** — Portal compromised, requiring IT intervention and delaying exam access.

**Emotional distress** — Significant stress, shame, and anxiety — reported in 89% of cybercrime victims.

**Loss of digital trust** — Arjun became anxious about all online platforms, impacting daily academic life.

### Legal Consequences (Indian Law)

| Section | Offence | Punishment |
|---|---|---|
| **Sec. 66C — IT Act 2000** | Identity theft using electronic means | **Up to 3 years imprisonment + Rs. 1 lakh fine** |
| **Sec. 66D — IT Act 2000** | Cheating by impersonation using a computer resource | **Up to 3 years imprisonment + Rs. 1 lakh fine** |
| **Sec. 420 — IPC** | Cheating and dishonestly inducing delivery of property | **Up to 7 years imprisonment** |

## 06 Prevention — What Should Have Been Done

**1 — CHECK THE SENDER EMAIL**
The display name said 'VIT Helpdesk' — but the actual domain was vitbhopal-helpdesk.in, not vitbhopal.ac.in. Always verify the full email address, not just the display name.

**2 — CHECK THE URL**
The fake site URL was visibly different from the official one. Always look at the address bar before entering any credentials.

**3 — NEVER SHARE OTPs**
No legitimate institution, bank, or government body will ever ask for an OTP via phone, email, or SMS. This is always fraud.

**4 — USE UNIQUE PASSWORDS**
Reusing the same password across platforms means one breach compromises everything. Use a password manager.

**5 — ENABLE TWO-FACTOR AUTH**
Even with a stolen password, 2FA blocks unauthorised access. This alone would have stopped the Gmail takeover.

**6 — CALL AND VERIFY**
When in doubt, call the institution using the official number from their website — not any number in the suspicious email.

| | |
|---|---|
| 7 | **REPORT IMMEDIATELY**<br>Report at cybercrime.gov.in or call Helpline 1930 (24x7). Early reporting helps shut down fake domains before more victims are affected. |

> ***Cybercriminals don't hack computers — they hack people.***
>
> ***Awareness is your strongest firewall.***

## 07 Sources and References


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